

Target Information and Profile System (TIPS): Risk Analysis and Management Strategies

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Abstract

This document addresses the risks associated with the development of Target Information and Profile System (TIPS). It proposes a methodology of development and some other processes to manage such risks including a continuous usability study and knowledge base validation.

Version Notice

All revisions made to this document are listed below in chronological order. The release number of this document does not relate to the release number of the application:

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Contents

1. Executive Summary	7
1.1 Objectives	7
1.2 Risk Identification	7
1.3 Risk Resolution Strategies	7
2. Introduction	9
2.1 Purpose	9
2.2 Author.....	9
2.3 Acknowledgments	9
2.4 Disclaimer	9
2.5 Opportunities and Benefits of TIPS.....	9
2.6 Scope	10
2.7 Assumptions and Dependencies	11
3. Risk Analysis and Management of TIPS	13
3.1 Risk Analysis and Assessment of TIPS.....	13
3.2 Possible Methods of Risks Management and Resolution	14
3.3 Risk Management Strategies and the Domains of Risk.....	15
3.4 Problem Domain Risk Analysis and Assessment	17
3.5 Problem Domain Risk Management - Spiral Development of TIPS	18
3.6 Problem Domain Risk Management: Usability Testing.....	20
3.7 Knowledge Base Risk Analysis and Assessment	21
3.8 Knowledge Base Risk Resolution: Knowledge Base Evaluation	21
3.9 Technology Risk Analysis and Assessment.....	22
3.10 Technology Risk Management: Technology Evaluation.....	23
3.11 System Interface and Integration Risk Analysis and Assessment	23
3.12 System Integration and Interface Risk Management: Resource, Quality Control and Testing.....	24
3.13 Knowledge Base Maintenance Risk Analysis and Assessment	24
3.14 Knowledge Base Maintenance Risk Management: Resource Identification and Process Engineering.....	24
3.15 Users' Culture Risk Analysis and Assessment.....	25
3.16 Users' Culture Risk Management: User Participation and Public Relation Lo- gistics.....	25
4. Issues	27
5. References	29
Appendices.....	31
Critical Quality Requirements of TIPS.....	31
Alternatives to Quality System and Control Process.....	31
Overall Quality System Design Philosophy.....	31
Mapping of the initial phase of TIPS to US WEST Development.....	32
Integration and Interface Risk Management: Quality Control Process Design.....	32
Document Notification	41

1. Executive Summary

1.1 Objectives

This document addresses the risks associated with the development of Target Information and Profile System (TIPS). It proposes a spiral evolutionary development methodology with built-in quality process to minimize those risks. It also recommends continuous knowledge-based system verification and validation tests and a series of system usability study plan as part of the development process.

1.2 Risk Identification

There are several classes of risks associated with TIPS - problem domain, knowledge base, technology, interface and culture:

1. problem domain - vague and not well-defined (modelling the strategies and knowledge of experts).
2. Knowledge Base - whether exploit knowledge can be acquired from experts or other sources to build the knowledge base in the time frame?
3. Interface and Integration - interface and integration with SSNQ (data services and user interface). Resource problem is one of the major source of risk associated with this area.
4. Technology - successful deployment of sales assistant type of expert system remains to be seen in U S WEST.
5. Culture - usually the most neglected but may be the most important factors in determining success of system.

1.3 Risk Resolution Strategies

1. TIPS Development Model - prototype-evaluation-production-feedback spiral development model to manage risks. A major check-point before production is used to minimize failure in the field. The spiral development model is used to extend the life-cycle and reduce the overall cost of the system. Because of our broadband strategies, many

new products will be expected to roll out in a few years time frame. We need a flexible development model and system to accommodate for changing products and market conditions.

2. Quality Control - quality control systems in areas of knowledge acquisition and external interfaces of TIPS (for example, data services and user-interface) which carry high degree of risks. Quality control systems will be designed to minimize extra effort to run the process and maximize self-autonomy of each area.
3. Rigorous but reasonable usability testing - establish an objective, statistically sound and realistic baseline for evaluating whether the recommendations generated are perceived as useful and acceptable by the end users. Besides, function and presentation usability tests will be conducted at various stages of the development process.
4. Risk-management tool - a risk analysis and assessment inventory of TIPS can be used to monitor and manage the risks associated with the project at regular time.
5. Knowledge-based system verification and validation tests - Amy Geoffroy from KBS and Rand Lewis from Testing will jointly develop a set of knowledge-based system verification and validation tests and apply them iteratively throughout the development of TIPS.
6. Involvement of in-house expertise from KBS in TIPS development - Amy Geoffroy from KBS has been involved in knowledge acquisition and the subsequent development of the knowledge base. She will coordinate and manage Carnegie Group Inc. in the knowledge engineering process.

Although TIPS makes use of advanced technology like knowledge-based system, the use of such technology is relatively novel in the domain of providing sales assistance within U S WEST. The applicability of the idea of providing sales assistance in the form of product recommendations from computer systems (regardless of technology) in the U S WEST environment still needs to be validated.

2. Introduction

2.1 Purpose

This document identifies the risks associated with the development of Target Information and Profile System (TIPS). It presents a set of risk management and resolution processes as possible strategies to minimize the risks. These processes include a risk analysis and management system, a spiral methodology of system development, a quality control system and a continuous usability evaluation process of TIPS.

The quality control process and usability evaluation test suites can be established to improve the quality of TIPS continuously throughout the life cycle of the system.

2.2 Author

The author of this document is Yat-Sang Hung, Member of Technical Staff, Human Factors and Standard Services, US WEST Advanced Technologies.

2.3 Acknowledgments

The author is grateful to the feedback and support of the following individuals - Judy Warner, Ed Barney, Gail Fontenelle, Chris Plott and Susan Wagner.

2.4 Disclaimer

The contents of this document represent only the views of the author. This document is intended to be used by Judy Warner, Ed Barney and Susan Wagner as a risk analysis reference document for TIPS from the perspective of the author.

2.5 Opportunities and Benefits of TIPS

TIPS is envisioned to be a system that will provide expert quality consultative sales advice to the CCEs through the use of expert system technology. The goals of TIPS are as follows:

- to foster customer loyalty as a strategic system in "The Gathering Storm" period of the competitive landscape [1]
- to position and prepare for the "Rough & Tumble" period by reducing cost and increasing service quality with SSNQ [1]

Successful development and deployment of TIPS will yield the following benefits:

- Consultative sales quality - the quality of product recommendations generated will approach or surpass recommendations by experts in consultative sales. The subject matter experts' (from CCEs, marketing and product management) knowledge of the customers and the market will be acquired, organized and distributed to less experienced CCEs as an intelligent sales assistant.
- Product recommendation consistency - product recommendations will be consistent across CCEs as they will be using a common source of knowledge. Such consistency is important in product management and development.
- Job enhancement - The system will reduce the mental load of CCEs so that they can devote more attention to customer rapport, interviewing and transitioning instead of straining themselves to generate product recommendations.
- Availability of salient customer information - TIPS will collect and store pertinent customer information for market analysis and research.
- Competitive advantage - the knowledge base of the system, if properly maintained and updated, will contain the most recently available knowledge of the products, the market and the needs of the customers drawn from the expertise of a diverse group of professionals (CCEs, marketing and product management). This "distributed" knowledge base may eventually perform better than individual subject matter experts.
- Interface to SSNQ - the integration and interface with SSNQ will create the synergistic effects of performing consultative sales and automating order entries with a common user interface.

2.6 Scope

This document will propose and initiate a set of proactive risk management and quality control processes to manage, monitor and improve continuously the development of TIPS in an evolutionary prototyping environment. It also proposes a series of prototype usability and knowledge base evaluation test to address to specific areas of risks.

In order to integrate quality and conformance of user requirements in the development process, a set of quality control processes specifically geared to the stages of prototyping and development is proposed. The quality control process is derived from some of the concepts and methods used in contemporary quality control systems like Total Quality Management [2]. The ISO 9000 Series of Quality management and quality assurance standards will also be used as a reference [3]. However, because of the unique situations and developmental constraints of TIPS, it may not be able to institute all the processes proposed in this document.

The risk management strategies described in this document serve to increase the likelihood of attaining success and reducing the overall cost of the project throughout its life-cycle. Given the complexities and intricacies involved, the very use of these processes does not necessarily imply any definite future success or cost reduction.

2.7 Assumptions and Dependencies

It is assumed that the knowledge acquisition phase of TIPS will finish its first iteration before December. The living laboratory for TIPS is scheduled to take place in March. Since Customer Service Records may not be available at that time, any tests on the system will be restricted to new connects.

3. Risk Analysis and Management of TIPS

3.1 Risk Analysis and Assessment of TIPS

There are several classes of risks associated with the development of TIPS. They are enumerated as follows:

1. Problem Domain - the problem domain is complex and vague. it is possible that TIPS might turn out to be a system that does not deliver what the users really need. Besides, user needs may change over time and may depend on many other factors that are beyond our control, like organizational and job requirements changes. To develop a successful system in this environment may involve evolutionary prototyping and continuous development to accommodate for new changes.
2. Knowledge Base - the subject matter experts may not be able to provide the necessary knowledge to generate accurate, substantive and non-trivial recommendations.
3. Technology - Although expert system technology has been used in various industries for several years and Carnegie Group Inc. is one of the foremost knowledge-based systems designers and providers in the field, the impact and success of expert system technology in the area of sales tips and assistance for CCEs within US WEST has yet to be proved.
4. Interface and Integration with SSNQ - the risk is medium initially within the time frame needed but should decrease dramatically once SSNQ developers can accommodate for the data services and user interface component of TIPS. At this point in time, it is elevated to high since qualified resources cannot be located to develop and integrate the GUI with SSNQ. There may be a trade-off between the amount of customer data collected and the usability of the system. There may also be a trade-off of the usability of the system under the design constraints of SSNQ.
5. Knowledge-based System Maintenance - the risk is medium to high if we do not establish a process to maintain the knowledge base. The knowledge base will need to be updated when new products roll out or market conditions and customer needs change.
6. Users' Culture - this is perhaps the most neglected aspect in system development but will play an important factor in the acceptance of the system. Studies of current expert systems show that the users tend to underrate the system if they are not involved in the system development process. On the contrary, developers and managers will tend to overrate the system. Other studies and evaluation of expert systems also show the importance of user participation. The risks associated with users' culture are not known and may depend on many other factors like organization changes, office objectives and job security issues.

Part of the risk analysis of TIPS has also been documented in TIPS: Recommendation and Evaluation of Future Direction [4].

3.2 Possible Methods of Risks Management and Resolution

Based on the above risk analysis and assessment, the following methods of risk management and resolution will be proposed to address the classes of risks above:

1. Problem Domain Risk Management - A development methodology is proposed to manage the risks in developing TIPS. This methodology is adapted from Boehm's approach to software risk management [10]. It can be used to extend the life-cycle of TIPS and reduce the overall cost of the system. User requirements can be iterated over different phases of the spiral. We need a flexible development methodology and system to accommodate changing products and market conditions. Specifically, a prototype usability study plan has been proposed to validate the user requirements.
2. Rigorous but reasonable knowledge base evaluation and usability tests - KBS and IT can establish guidelines for evaluating whether the recommendations generated are accurate, valid and non-trivial. The method of evaluation can be communicated to the vendor clearly and early in the process so that they can work towards the conformance of user requirements. They can design knowledge-based system verification and validation tests to evaluate TIPS accordingly. In the usability testing, we can establish an objective, realistic and statistically sound base line for measuring whether the recommendations are perceived as useful or acceptable by the end users across different regions and cities. Other benefits of this methodology include exploration of regional differences among experts and input to the next phase of the spiral to improve the quality of the system.
3. Technology Awareness Program - in order to gain a competitive edge for the system, we can explore the application of new technologies to the problem domain periodically. Other useful technologies can be embedded into the system to add value to the system (for example, multimedia, neural network...).
4. Quality Control Systems - the high-risks components of the system (for example, the external interfaces of the system) can be designed and implemented with the use of quality control systems. Examples of such interfaces and the associated processes include knowledge acquisition process ("interface" between knowledge base and subject matter experts), data services interface with SSNQ, user interface of TIPS conforming to the model used in SSNQ... The quality control system will be designed to minimize extra efforts to run the quality process and to maximize autonomy in the respective domains of the developers.

5. Understanding of Users' Culture, User Participation and Public Relation Logistics - it is well known in expert system research that the more users participate in the development process, the more they will accept and use the system. We can make use of every opportunity to gain users' acceptance. The debriefing of the knowledge base evaluation, for example, can also be used to acquaint with the overall objective of TIPS and encourage them for input. Attention can be made to improve our understanding of the concerns of the users and impact of the system on their culture. This has important implications in their training as well as the formulation of business office policy and objectives.
6. KBS Involvement - the expertise of KBS within U S WEST can be utilized in all phases of the development of TIPS. Amy Geoffroy will coordinate and manage the knowledge engineering efforts of CGI as well as design the knowledge-based system verification and validation tests of TIPS.

3.3 Risk Management Strategies and the Domains of Risk

The following table shows the mapping of the risk management strategies to different domains of risks. Quality and risk control systems can be utilized across all the risk domains as the framework and are not shown below:

Risk Domain	Critical Period	Risk Management Strategies
Business Problem	User Requirement and Prototype Stage Pre-Deployment Stage	1. User and Management Involvement (CCEs, marketing, executive sponsor) - TIPS Development Team 2. Usability Study of major functions of TIPS to validate user requirements - Human Factors 3. Knowledge Usability Study to validate the usefulness and acceptability of the recommendations to end users - Human Factors 4. Spiral Risk-Based System Development - TIPS Development Team

Table 1: Risk Domain and Risk Management Strategies

Risk Domain	Critical Period	Risk Management Strategies
Knowledge Base	Knowledge Acquisition and Prototype Stage Deployment Stage	1. Identification of SMEs from marketing, product management and CCEs - SSNQ Client Project Manager 2. Cross validation of knowledge across different SMEs - Knowledge Engineers 3. Knowledge-based system verification and validation - Knowledge Engineers 4. User Involvement in building a natural model for the business domain - Knowledge Engineers 5. Consistent training of end users with the model used in the system - Training and Documentation
Technology	Prototype Stage to System Deployment	1. Review of different knowledge-based system technologies at appropriate time - TIPS Development Team 2. Knowledge-based System Verification and Validation - Knowledge Engineers and Testing Engineers
Presentation	Prototype Stage to System Deployment	1. Prototype Usability Study - Human Factors Engineer 2. User involvement in UI design - Human Factors Engineer
Integration and Interface with SSNQ	Prototype Stage to System Deployment	1. Identification of qualified resource for development - SSNQ Technical Project Manager 2. System testing - Testing Engineer

Table 1: Risk Domain and Risk Management Strategies

Risk Domain	Critical Period	Risk Management Strategies
User Culture	User Requirement Stage to Post System Deployment	1. User involvement in system development - TIPS Development Team, Client Organization 2. Demonstration and feedback sessions whenever there is a chance - TIPS Development Team, Client Organization
Organization	User Requirement Stage to Post System Deployment	1. Formulation of business office policy without conflicting goals - Client Organization
Knowledge Base Maintenance	Post System Deployment	1. Identification of qualified resource for knowledge base and expert system maintenance - Client Organization

Table 1: Risk Domain and Risk Management Strategies

3.4 Problem Domain Risk Analysis and Assessment

The overall objective of TIPS is to organize, represent and present sales, product and marketing knowledge in a form that can be shared, collected, communicated and utilized effectively by Customer Contact Employees and other departments involved [5]. In the initial design and implementation, TIPS is envisioned to be an intelligent sales tool supporting consultative sales activities of CCEs. It will be integrated with Sales and Service Negotiation and Quality (SSNQ) System and generate customized product recommendations based on the telecommunication needs of the customers.

One of the questions arising from the user requirement collection process is whether a tool that generates recommendations fulfills the primary needs of the users. From the focus group sessions of the CCEs, it was found that customer information that supports the recommendations of the CCEs is among the most desired functions of a sales assistant [4,6]. Part of the information will be collected and stored as customer profile or derived from existing data in the legacy systems. However, some of the critical information like actual usage patterns may not be available in this phase of

the project. The concept of an expert system giving sales tips is embraced by the management and the marketing staff. It remains to be seen whether it will be a valid and usable system for operational users.

There are published guidelines for conducting consultative sales [7]. However, the actual contacts are highly individualistic. The flow and the depth of each contact depends on situational factors (like the nature of call, accessibility of the business office at that time) as well as individual personality and motivations. In this sense, the problem domain is relatively ill-defined and depends on the complex interactions of many variables.

Another source of risk is the complexities of the product recommendations required and the difficulties to give recommendations on a per telephone line basis. It is not clear whether such detailed analysis and recommendations are possible without extensive and time consuming consultation. This will present difficulties in designing an efficient human-computer interface for CCEs.

Given the circumstances above, the risk of developing a system that may miss the problem domain is medium to high if we do not evaluate and contain such risks at appropriate time.

3.5 Problem Domain Risk Management - Spiral Development of TIPS

A variant of Boehm's Spiral Model of Development can be used to structure the development process of TIPS [10]. A steering committee and various quality circle groups can be formed to oversee the development effort. This model is an attempt to formalize and establish process control at different stages of development. The model organizes the life cycle of TIPS into a series of phases. Each phase consists of the following stages or processes:

1. Prototyping stage - activities include user requirement collection, knowledge acquisition, high-level design, system architecture design, functional specification and user interface specification. These tasks culminate in the delivery of a functional prototype system (depending on the availability of SSNQ data services and resources). Particular quality consideration will be given to knowledge acquisition, which is the bottleneck of the stage and the interface with SSNQ (data services and user-interface design). A prototype usability study has been designed to address the validity of the user requirements in the initial phase[11].
2. Evaluation and review stage - knowledge base evaluation test and usability testings [11] will be conducted at different stages of the project from panels without behavior to fully functional prototype
3. production and deployment stage - the functional prototype goes into production and deployment stage after the knowledge base evaluation, the usability study and the software testing (if needed) pass the

acceptance/certification review. After the delivery of the production system, IT Testing Engineers will perform system and load testing of the system. This process will help to ensure that a quality system will be deployed in the field.

4. feedback and review stage - surveys and usability studies will be conducted on the users and maintainers of the system several months after deployment. At that time, users and maintenance personnel will be more familiar to the system to give accurate evaluations and feedbacks.

The details of a possible implementation scheme is available at the appendix.

The following table summarizes the activities associated with different stages of development in a phase of the spiral development process:

Stage	Resource	Input	Activities	Deliverable
Prototype Process	TIPS Development Team	User requirements (Current Iteration)	User requirement collection	User Requirements Document
		SMEs input	Knowledge acquisition	System Architecture Document
		KB evaluation study (from previous phase)	Functional specification	Functional Specification Document
		Usability study (from previous phase)	User interface design	User Interface Specification Document
		Feedback Study (from previous phase)	Prototype Implementation	Usability Studies Plan
				KB System Verification and Validation Plan
				Functional Prototype

Table 2: Spiral Development Model of TIPS

Stage	Resource	Input	Activities	Deliverable
Evaluation and Review Process	Knowledge Engineer	User Requirements	KB Evaluation	KB Evaluation Report
	Human Factors Engineer	Functional Specification	Usability Studies	Usability Report
	Software Testing Engineer	Prototype Usability Testing Plan	Software Testing	Software Testing Report
	Steering Committee	UI Specification		
Production and Deployment Process	TIPS Development Team	Functional Prototype	Production of TIPS	Production Model of TIPS
	TIPS and SSNQ Deployment Team	Reports from Previous Process	Deployment and Training	Deployment and Training
Feedback and Review Process	Knowledge Engineer	Usability Studies and Software Testing	Usability Studies	Feedback Report
	Human Factors Engineer	Results from Evaluation Process	Feedback and Survey	Phase Review Report
	Software Testing Engineer			
	Steering Committee			

Table 2: Spiral Development Model of TIPS

3.6 Problem Domain Risk Management: Usability Testing

After the delivery of the alpha version of the prototype of TIPS, it can be tested against the operational users (CCEs) for function and presentation usability. The function usability test will focus on the question whether the operational users really need and use the recommendations provided by the system. The presentation usability test will address whether the screen layout and flow can facilitate consultative behavior. The testing can be done with the coming iteration of usability testing of the prototype of SSNQ/TIPS.

Details of the draft usability studies plan have been designed and released on September 15, 1993.

A more rigorous approach will be employed to assess the usability of the sales advice of TIPS for operational users across three regions as well as marketing and product management staff before deployment. Besides, a semi-structured usability evaluation of the knowledge acquired from knowledge acquisition sessions may be validated against the CCEs in the usability iteration of SSNQ/TIPS. Such tests will be focused on the usability of the sales tips and recommendations generated from the system regardless of what technology is used to design and implement it.

3.7 Knowledge Base Risk Analysis and Assessment

Knowledge Base Risk Analysis has been extensively covered in the "TIPS: Recommendation and Evaluation of Future Direction" under Problem Domain Analysis (section 4.5 p.19-22) [4]. The subject matter experts may not be able to provide the necessary knowledge to generate accurate, substantive and non-trivial recommendations. This possibility has been identified in the user requirement collection process. Mitch Glasser of Strategic Marketing also made a similar point in the conference call on June 8, 1993. His research in the project DARWIN, suggests that a capability-based classification of products to match the needs of customers may be useful as the first step in representing consultative sales knowledge of CCEs. He contends that their efforts are preliminary and much work has to be done before the knowledge becomes suitable for expert system implementation. His conception of capability-based categories and hierarchy of needs seems to be one of the good candidates for the representation and organization of knowledge for the expert system. Other suggestions include a product parameter model in terms of customer needs and a customer-centered model of goals, strategies and needs as well as a model of customer business functions and profile [8,9]. Since knowledge acquisition is generally regarded as the bottleneck of the knowledge base building process and knowledge of the SMEs has yet to be elicited, risk is scored as medium in this area.

3.8 Knowledge Base Risk Resolution: Knowledge Base Evaluation

The various SMEs from Marketing and Product Management have been identified and arranged by Ed Barney. They are highly regarded experts in the area. They also possess organized knowledge necessary for the problem domain. The availability of the experts (together with 2 client leads from the CCEs) should substantially reduce the risks of knowledge acquisition, provided that we can conduct the knowledge acquisition process in a systematic and structured way.

Since the capability-model may take time to understand (from the CCEs and customers perspective) and in fact is not recommended by Mitch to be included in the sales tool, it is important to use a common-sensical and natural model of the domain that is easily communicated and understood by the CCEs, marketing and product development as well as the customers. It can also serve as a common basis for validating the knowledge as well as training the CCEs to use the system [9]. Without a natural and intuitive model of the domain, the users may find it difficult to use and accept the system.

Continuous testing of the knowledge base throughout the development process has been recommended [11]. SMEs from Marketing, CCEs and Product Management can rate scenarios generated from the prototype in the knowledge acquisition and knowledge base building stage. They can also validate knowledge through the use of a common model or other means as determined by Amy Geoffroy and Rand Lewis.

Details of the knowledge-based verification and validation plan will be designed primarily by Amy Geoffroy and Rand Lewis with my input and suggestions.

3.9 Technology Risk Analysis and Assessment

Knowledge-based system is a proven technology that can provide expert advice if we constrain the system to a limited and restricted domain. The problem domain of TIPS, on the other hand, is fairly complex as the recommendations will be generated on a per telephone line basis. It will be challenging to develop a substantive expert system that can handle such situations.

We have selected Carnegie Group Inc. as technology provider for the development of expert system, the risks associated with technology will be linked to their capability of delivering a quality expert system engine (and knowledge base). Amy Geoffroy from KBS will be our in-house expertise to evaluate the internal quality of their engine and the knowledge base they develop. Over the years as a strategic partner of US WEST, Carnegie Group Inc. has successfully developed and deployed several expert systems. These include ARMAR (deployed in 14 states and resolves 3 types of RMAs), CSN (used by BGS, Federal and IEC market units and covers major ISDN orders) and several help desks.

Since there is no sophisticated knowledge-based system providing sales assistance at this point and the proposed system is relatively novel in its domain of operation, it is difficult to assess the effect of technology on the project. After all, the use of advanced technology does not necessarily

warrant the success of a system, even though it addresses and solves the business problem. User culture will play a very important part in the successful deployment and utilization of a system.

The risk associated with successful development and deployment is high at this stage of the project. The active role and participation of KBS will help to minimize the risks.

3.10 Technology Risk Management: Technology Evaluation

At the beginning of each phase, a technology review and evaluation can be conducted to explore new technology opportunities and uses. It can be conducted in relation to the specific requirements of TIPS for possibilities of enhancements.

In the initial phase, the expert system technology will be evaluated by the quality of the product recommendations it generates. Evaluation will be centered on the effectiveness of the technology to give quality recommendations and the chain of reasoning why specific sets of recommendations are generated. Because of the relative shortage of in house expertise in expert system development and the lack of relatively objective evaluation tool of knowledge-based systems, we will focus on the knowledge base validation and its interplay with the engine. Further testing will be conducted by knowledge engineer from KBS and software testing engineers of IT as appropriate.

3.11 System Interface and Integration Risk Analysis and Assessment

TIPS is dependent on SSNQ for interface and integration. Since SSNQ has a tight schedule and software integration needs to be done in the architecture level as well as at the functional and data services level, it is anticipated that more resources have to be identified and brought into the SSNQ development team to accommodate such changes. As of September 30, 1993, no definite answers have been given by SSNQ to TIPS development team as to what functionalities they can develop for TIPS in the initial phase. Some of the architecture decisions have not been made by SSNQ, it is difficult to design data service interface for the engine of TIPS at this point in time.

The knowledge base building and the expert system engine by itself are designed in a modular manner independent of these interfaces and are on schedule. The risk of interfacing with SSNQ at this point is high. Since SSNQ cannot identify appropriate resources as of September 30, 1993, the schedule for the delivery of TIPS will probably slip. Meanwhile, CGI can proceed with the knowledge acquisition (in collaboration of US WEST KBS) and the design of the engine.

The interface design of TIPS to SSNQ may be also be constrained by the pre-existing architecture of SSNQ. This may become more apparent as the project moves on.

3.12 System Integration and Interface Risk Management: Resource, Quality Control and Testing

One of the pressing needs in this area is the identification of qualified resource to support user interface integration and coding from IT before other risk management techniques can be applied. The integration of the architecture of TIPS with SSNQ also requires a time and resource commitment from SSNQ engineers to incorporate the model/view/controller paradigm into TIPS as well as to provide an application programming interface for the data services of SSNQ.

After resources are identified and committed, a quality control system (or a protocol to address and resolve issues) can be put into place in the area of user interface and data service integration similar to the protocol used by SSNQ or that described in the appendix.

The TIPS development team as well as the testing engineers of IT will also conduct tests on integration with SSNQ at appropriate time.

3.13 Knowledge Base Maintenance Risk Analysis and Assessment

Constant maintenance of the knowledge base will be required for new products, changes in the demographics and needs of customers as well as the market situations. Personnel have to be identified to these tasks. These may include a knowledge engineer from KBS for periodic maintenance and enhancement of the rules and the engine, and at least a staff member from SBS to coordinate and manage the changes. The risk is relatively low at this time. However, its status should remain unresolved until appropriate personnel have been identified.

3.14 Knowledge Base Maintenance Risk Management: Resource Identification and Process Engineering

Qualified resources from KBS and SBS can be identified at a later stage of the project to maintain and update the knowledge base. Processes of knowledge base maintenance can be jointly designed by KBS and SBS. At this point in time, Darlene Warren, Cathe Valdez, Patti Waddell and Amy Geoffroy can plan and further develop the required processes.

3.15 Users' Culture Risk Analysis and Assessment

The purpose of an expert system is to make consultative sales knowledge available to every CCE at their desk. The system may change the job structure and flow of the CCEs. It is natural that some users may not be too ready to accept the system. Such concerns can be seen in the focus groups of the CCEs in the user requirements collection process. One expert CCE, for example, said that the system should not be designed in such a way that everyone can use it. Another CCE remarked that the system should not turn them into "robots".

It is difficult to assess the risk in this area as the CCEs may not be aware of the full implications of a knowledge-based system until deployment. The risk in this domain is medium to high.

3.16 Users' Culture Risk Management: User Participation and Public Relation Logistics

It is well known in expert system research that the more users participate in the development process, the more they will accept and use the system. We can make use of every opportunity to gain users' acceptance. The debriefing of the knowledge base evaluation, for example, can also be used to acquaint the users with the overall objective of TIPS and encourage them for input. Attention can be made to improve our understanding of the concerns of the users and impact of the system on their culture. This has important implications in their training as well as the formulation of business office policy and objectives.

Since the development and deployment of expert system may imply changes in job structure and flow, the system may be met with user resistance and even resentment. This may constitute a serious problem in the business offices. Various questions regarding with the role of the expert system and the possible impact on the nature of the job have been raised by the CCEs participating in the focus groups. Although the reactions of the CCEs towards TIPS seem to be positive in those groups, it is difficult to project their reactions when the system is developed and deployed. In effect, the deployment and use of TIPS cannot change the culture of the CCEs per se. TIPS may be able to facilitate some changes but should not be viewed and used as a major tool to change the behavior of CCEs.

The assimilation (or accommodation) process of CCEs to a customer-centered sales consultative culture depends on multiple and complex factors such as business office policy, company vision, market environment and information system support. It also depends heavily on behavior drivers like compensation and job evaluation criteria. Without a consistent management objective and strategy, it may be difficult to achieve the vision of the company.

4. Issues

List and describe any points or matters of discussion, debate, or dispute that impact the project that have not been resolved.

Note: This is a working, dynamic section throughout the life of the project, including the preparation of this document. Use this section as a repository of all the things you do not know, things you must research, and remove them when you have resolved them.

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Appendices

The following is a possible implementation plan of the suggestions made in this document. It is noted that many other variations and modifications to the following process are possible to achieve the same goals. The use or modifications of these processes depend on circumstances.

Critical Quality Requirements of TIPS

The critical quality goals as stipulated by the user requirements of TIPS are as follows:

- TIPS will generate accurate, valid and non-trivial recommendations from customer needs and profile information.
- TIPS will accommodate changes in market conditions, pertinent customer demographics and product sets without massive redesign or redevelopment.
- The user interface of TIPS should be constructed optimally to allow CCEs to collect salient customer information under time and environmental constraints.

Alternatives to Quality System and Control Process

Because of the time and resources involved and the tight schedule, a monthly core team meeting and a periodic update and report of progress and issues from various teams (weekly to monthly, depending on circumstances) can be used in lieu of some of the quality control processes described below.

The core team meetings we had in August and September were effective in addressing and resolving important issues and questions. Given the success of these meetings and the progress of the project, it may not be necessary to institute the entire process described below. A partial implementation of the process will probably be adequate to ensure quality without straining resources.

Overall Quality System Design Philosophy

The goals of conformance of user requirements and integrating quality into the system in the design and prototyping phases can be attained by the following process:

1. TIPS life-cycle process definition and description - the risk management and quality process should encompass the life-cycle of TIPS; it serves to optimize the balance between the total cost and the value TIPS can add to the business throughout the life-cycle. The definition and description of TIPS life-cycle process will form the substratum on which quality systems can be built upon.
2. TIPS quality system definition and calibration - the specification of the quality system is geared to each stage and phase of the TIPS life-cycle process. The calibration of the system serves as benchmarks on which further evaluations of the system can be based.
3. Continuous quality control and iterative improvement - the purpose of continuous quality control is to bring the risk associated with the development to a minimal. It serves to prevent the occurrence of major quality problems which will be very expensive to fix.
4. Critical evaluation and usability studies at different stages - these studies will form part of the basis to assess future risks of development and the general utility of TIPS. If it cannot fulfill its success criteria, different action plans can be invoked to handle the situations.

Mapping of the initial phase of TIPS to US WEST Development

The following table shows the mapping of the initial phase of the development of TIPS to US WEST Project Lifecycle Methodology:

US WEST Project Lifecycle	TIPS PEPF Spiral
Project Initiation	Prototype Stage
System Requirements	Prototype Stage
System Engineering	Prototype Stage, Evaluation and Review Stage
System Development	Evaluation and Review Stage, Production and Deployment Stage
Deployment	Production and Deployment Stage

Table 3: Initial Phase of TIPS

Integration and Interface Risk Management: Quality Control Process

Design

A. Criteria of Quality Control Process of TIPS

Because quality control is a continuous process and extends to cover the life cycle of TIPS, it will be designed with the following criteria to avoid possible dislocation or interruption of the quality system:

- technology independence - the quality control process should be independent of the technology that TIPS uses. TIPS may integrate or utilize new technologies in the future and the quality control system should be flexible and general enough to cover these circumstances.
- role-based owner participation - the success of any quality control system depends on the proactive participation of the owners of the process and the subprocesses. Team members will determine and agree on the agenda, objectives, methods and procedures specific to their processes.
- owner autonomy and independence - since process owners know their processes best, they will design and define their control procedures and measurements from their experience and the input of their peers. They should document the process clearly and concisely so that the procedures can be followed, reviewed or modified independent of owners. In this way, personnel change will have little impact on the running process.
- critical- success-parameter-driven process - the quality process will be driven by goals as defined in section 4.4 as well as the user requirement document of TIPS. These parameters may change in later phases and in general depends on the iteration of user requirement at that phase.
- structuration of process control - a steering committee for the overall TIPS process and other quality circles for its sub-processes will be established.
- procedure formalization - the major procedures for prototyping and evaluation can be formalized for consistency.
- change administration - changes in personnel, procedures and objectives are inevitable. The design of all the processes and evaluation plans will attempt to accommodate such changes as smoothly as possible.

B. Structure of Quality Control System of TIPS

A steering committee for the quality of TIPS can be formed to oversee the quality process in each phase of development of TIPS. The committee will only convene as needed to resolve major quality issues or decisions. It must be stressed that this is only a suggestion and many variations are possible. The members of the committee can consist of the following members:

1. Client Project Manager of SSNQ
2. Technical Project Manager of SSNQ

3. Technical Lead of SSNQ
4. Client Project Lead of TIPS (SME)
5. Human Factors Engineer of TIPS
6. SSNQ Client Lead (SME)
7. TIPS Knowledge Engineer (KBS)
8. TIPS Vendor Representative (Knowledge Engineering)
9. TIPS Development Team Representative
10. TIPS Client Marketing Representative (SME)
11. SSNQ/TIPS Testing Engineer
12. SSNQ/TIPS Client Training Specialist

The following Quality Circle Committees can be established:

Knowledge Acquisition Quality Circle

Participants:

1. Knowledge Engineers from US WEST and Vendor
2. Client Project Manager of TIPS (SME)
3. SSNQ Client Lead (SME)
4. TIPS Client Marketing Representative (SME)
5. TIPS Client Product Management Representative (SME)

User Interface Quality Circle

Participants:

1. UI Engineer (US WEST)
2. UI Engineer (Vendor)
3. Human Factors Engineer of TIPS
4. TIPS Client Project Lead (SME)

5. SSNQ Client Lead (SME)
6. Human Factors Engineer(s) of SSNQ

Database Interface Quality Circle

Participants:

1. Database Engineer (US WEST)
2. Database Engineer (Vendor)

C. Measurements of Quality and critical success parameters of TIPS

The goals of conformance to user and quality requirements can be transformed into the following measurements taken at different time intervals (for example, day, week, month, stage or phase) throughout the process. The measurements described below can be adopted or modified in the knowledge-based system verification and validation plan as decided by the authors of the plan.

The following table organize and map the quality requirement goals in the initial phase to each type of measurements by time and purpose:

Goal	Measurement	Time
<i>valid recommen-</i> <i>dations</i>	rating of validity of reasoning	Knowledge Vali- dation Sessions Internal Proto- type Testing Knowledge Base Evaluation Post-Deployment Survey

Table 4: Mapping of Quality Goals to Measurements

Goal	Measurement	Time
<i>accurate</i> recommendations	accuracy scores acceptability scores	Knowledge Validation Sessions Internal Prototype Testing Knowledge Base Evaluation Post-Deployment Survey
<i>non-trivial</i> recommendations	distribution of products recommended	Knowledge Validation Sessions Internal Prototype Testing Knowledge Base Evaluation Post-Deployment Survey
<i>effective</i> user interface	survey, attitude scale	Internal Prototype Testing Prototype Usability Study Usability Study (Living Lab.) Post-Deployment Survey

Table 4: Mapping of Quality Goals to Measurements

D. Prototype Process Quality Control

The Prototype Process can be organized into several distinct but possibly overlapping sub-processes:

Sub-process	Owners	Input	Quality Process	Output
User Requirements	TIPS core team	SBS Management SBS Marketing Strategic Marketing SMEs	Critical Review Feedback	User requirements Document
High-Level Design Specification	TIPS Vendor and IT	User requirements Document	Critical Review Feedback	High-Level Design Specification Document
Knowledge Acquisition and Engineering	Knowledge Engineers SMEs	SMEs from CCEs, Marketing and Product Management	Knowledge Validation Sessions	Knowledge Base Specification
System Architecture	System Developers (IT and Vendor)	User requirements Document System Architecture Document of SSNQ	Critical Review Feedback	System Architecture Document
Functional Specification	System Developers (IT and Vendor)	User Requirements Document	Critical Review Feedback	Functional Specification Document

Table 5: Prototype Stage Quality Control Process

Sub-process	Owners	Input	Quality Process	Output
User Interface Specification	Human Factors Engineer UI Engineers (IT and Vendor) TIPS Client Project Lead SSNQ Client Lead	User Requirement Document SME	Review by SMEs Review by Human Factors Engineers of SSNQ	User Interface Specification Document
Low-Level Design Specification	System Developers (IT and Vendor)	High-Level Design Specification Document	Design Reviews Structured Walkthrough	Low-Level Specification Document
Implementation and Coding	System Developers (IT and Vendor)	Documents as stated from above	Continuous Testing	Functional Prototype

Table 5: Prototype Stage Quality Control Process

E. Evaluation and Review Process Quality Control

The Evaluation and Review Process can be organized into several distinct but possibly overlapping sub-processes:

Sub-process	Owners	Input	Quality Process	Output
Quality and User Requirements Analysis	Client Project Lead SSNQ Client Project Lead SSNQ Client Project Manager	User requirements of TIPS	Input from SMEs of Client Organization Critical Review	TIPS Success Criteria and User Acceptance Criteria
Knowledge Base Validation and Evaluation Design	Knowledge Engineer (KBS) and Testing Engineer (IT)	Quality and User Requirements	feedback from clients and core team of TIPS	Knowledge Base Verification and Validation Plan
Usability Studies and Evaluation Design	Human Factors Engineer	Quality and User Requirements Usability Studies Outline	feedback from clients, vendor of expert system and peer human factors engineers	TIPS usability studies Plan
Knowledge Base Evaluation	Knowledge Engineer (KBS) and Testing Engineer	Knowledge Base Evaluation Plan	Controlled Testing Conditions	Knowledge Base Evaluation Report
Knowledge Base Evaluation Review	Steering Committee or SSNQ Client and Technical Manager	Knowledge Base Evaluation Report	Critical Review	Decision of Acceptability
Usability Studies	Human Factors engineer	Usability Studies Plan	Controlled Testing Conditions	Usability Studies Report

Table 6: Evaluation and Review Quality Control Process

Sub-process	Owners	Input	Quality Process	Output
Usability Studies Review	Steering Committee or SSNQ Client and Technical Manager	Usability Studies Report	Critical Review	Decision of Acceptability
Software Testing and Verification (if needed)	Testing Engineer	User Requirements Functional Specification Document	Controlled Testing	Software Testing Report
Overall Evaluation Review	Steering Committee or SSNQ Client and Technical Manager	Prototype Usability and Knowledge Base Evaluation Reports Software Testing Report (if needed)	Critical Review	Decision of Future Direction

Table 6: Evaluation and Review Quality Control Process

F. Acceptance Process/Plan of TIPS

The Client Project Manager and Technical Project Manager of SSNQ as well as the Client Organization will decide the criteria to assess the knowledge base as well as the usability of the system.

Document Notification

The individuals listed below received a copy of this document on *Oct 7, 93*.

TIPS: Risk Analysis and Management Strategies

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